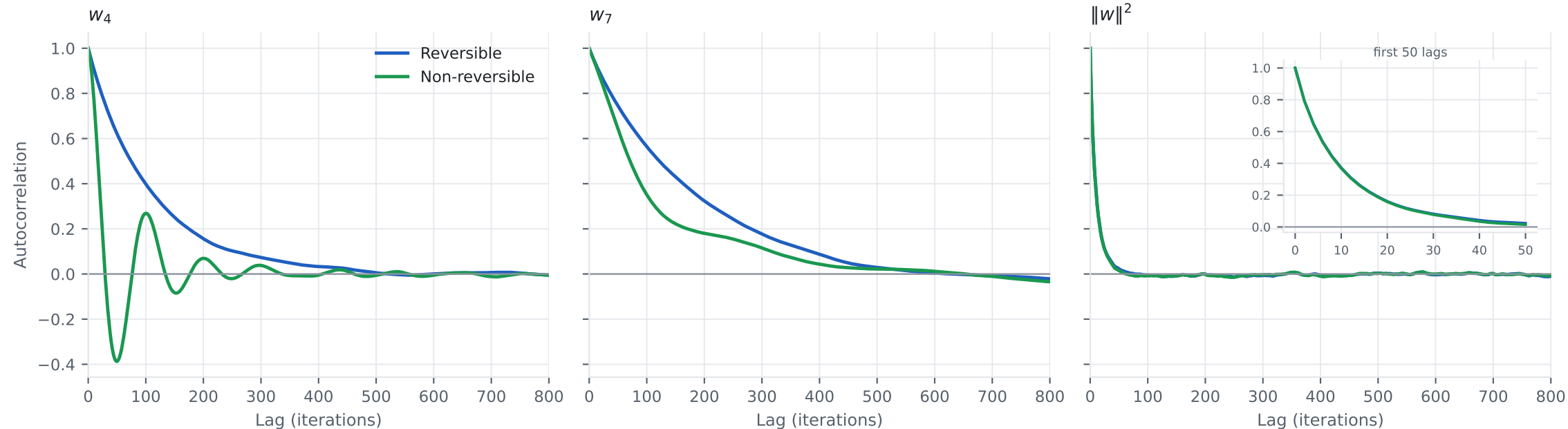


Why: the non-reversible chain decorrelates sooner — except where it does not



Mean normalised autocorrelation over $R = 48$ coupled chains, Titanic ball, $\eta = 0.0001$, $s = (5,5,5)$, 12000 iterations, 40% burn-in, seed 6100. w_4 is the coordinate with the largest gain (x5.5 in ESS per second), w_7 the slowest-mixing one and the smallest gain, and $\|w\|^2$ the one test function that mixes slightly WORSE under the non-reversible dynamics -- the inset shows the two curves are visually indistinguishable there, the x0.91 being too small to see. The area under each curve is what the integrated autocorrelation time measures. The oscillation in w_4 is the signature of the rotation: J turns the drift, so the chain circulates instead of diffusing back and forth, and the correlation crosses zero within ~ 50 iterations instead of decaying over ~ 300 .